

The properties of logarithm established in 33–35 are used in sections 11.4 and 11.5.

7.2.34. Prove that for all positive real numbers a , x , and y with $a \neq 1$,

$$\log_a(xy) = \log_a x + \log_a y$$

(Note: Substituted subscript b for a)

Proof:

Suppose a , x , and y are positive real numbers, with $a \neq 1$.

We must show that $\log_a(xy) = \log_a x + \log_a y$.

Let $a, x, y \in \mathbb{R}^+$, $a \neq 1$

Definition of Logarithm and Logarithmic functions:

$$\log_n r = s \leftrightarrow n^s = r$$

Let $z, u, v \in \mathbb{R} \mid u = \log_a x, v = \log_a y$, and $\log_a(xy) = z$, then:

$$\log_a(xy) = z \leftrightarrow a^z = xy$$

$$\log_a x = u \leftrightarrow a^u = x$$

$$\log_a y = v \leftrightarrow a^v = y$$

Hence, $a^z = xy$, $a^u = x$ and $a^v = y$. Then:

$$xy = a^u a^v = a^{u+v}$$

$$xy = a^{u+v}$$

$$xy = a^z$$

$$a^z = a^{u+v}$$

Therefore, $z = u + v$, and since $z = \log_a(xy)$, $u = \log_a x$, and $v = \log_a y$, then $\log_a(xy) = \log_a x + \log_a y$, which was to be shown.

□

Solution: for all positive real numbers a , x , and y with $a \neq 1$, $\log_a(xy) = \log_a x + \log_a y$ is true.